LAPORAN PRAKTIKUM

INTERNET OF THINGS II



Di susun oleh:

FAIZ HISYAM RASYID HARAHAP (25782068) TRI – 3C

Dosen Pengajar :

Dr. Ir. Septafiansyah Dwi Putra, S.T., M. T.

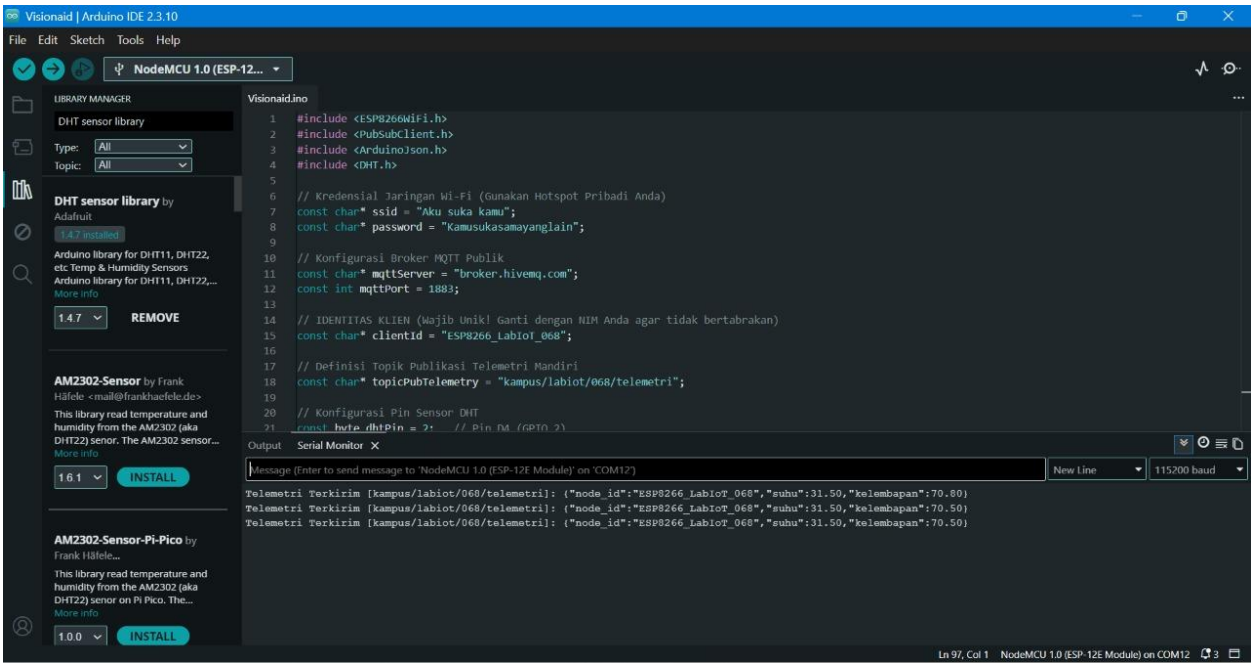
Agus Ambarwari, S.Pd., M.kom

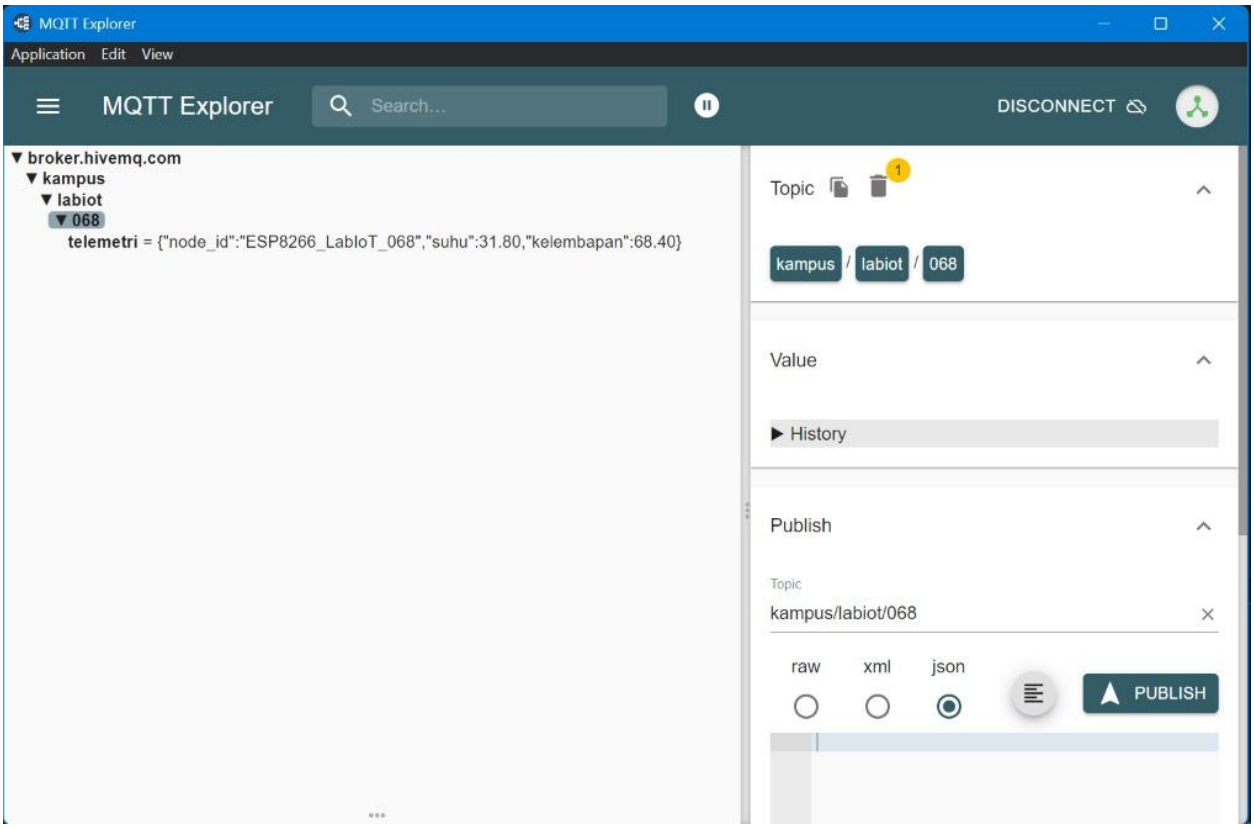
TEKNOLOGI REKAYASA INTERNET

JURUSAN TEKNOLOGI INFORMASI

POLITEKNIK NEGERI LAMPUNG

Praktikum 1


Visionaid | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
LIBRARY MANAGER
DHT sensor library
Type: All
Topic: All
DHT sensor library by Adafruit
1.4.7 installed
Arduino library for DHT11, DHT22, etc Temp & Humidity Sensors
Arduino library for DHT11, DHT22,...
More info
1.4.7 REMOVE
AM2302-Sensor by Frank Häfele <mail@frankhaefele.de>
This library read temperature and humidity from the AM2302 (aka DHT22) senor. The AM2302 sensor...
More info
1.6.1 INSTALL
AM2302-Sensor-Pi-Pico by Frank Häfele...
This library read temperature and humidity from the AM2302 (aka DHT22) senor on Pi Pico. The...
More info
1.0.0 INSTALL
Visionaid.ino
#include <ESP8266WiFi.h>
#include <PubSubClient.h>
#include <ArduinoJson.h>
#include <DHT.h>
// Kredensial Jaringan Wi-Fi (Gunakan Hotspot Pribadi Anda)
const char* ssid = "Aku suka kamu";
const char* password = "Kamusukasamayanglain";
// Konfigurasi Broker MQTT Publik
const char* mqttServer = "broker.hivemq.com";
const int mqttPort = 1883;
// IDENTITAS KLIEN (Wajib Unik! Ganti dengan NIM Anda agar tidak bertabrakan)
const char* clientId = "ESP8266_LabIoT_068";
// Definisi Topik Publikasi Telemetri Mandiri
const char* topicPubTelemetry = "kampus/labiot/068/telemetri";
// Konfigurasi Pin Sensor DHT
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM12')
New Line
115200 baud
Telemetri Terkirim [kampus/labiot/068/telemetri]: {"node_id":"ESP8266_LabIoT_068","suhu":31.50,"kelembapan":70.80}
Telemetri Terkirim [kampus/labiot/068/telemetri]: {"node_id":"ESP8266_LabIoT_068","suhu":31.50,"kelembapan":70.50}
Telemetri Terkirim [kampus/labiot/068/telemetri]: {"node_id":"ESP8266_LabIoT_068","suhu":31.50,"kelembapan":70.50}
Ln 97, Col 1 NodeMCU 1.0 (ESP-12E Module) on COM12


MQTT Explorer
Application Edit View
MQTT Explorer
Search...
DISCONNECT
broker.hivemq.com
kampus
labiot
068
telemetri = {"node_id":"ESP8266_LabIoT_068","suhu":31.80,"kelembapan":68.40}
Topic
1
kampus / labiot / 068
Value
History
Publish
Topic
kampus/labiot/068
raw xml json
PUBLISH

Praktikum 2

sketch_oct19b | Arduino IDE 2.3.10

File Edit Sketch Tools Help

NodeMCU 1.0 (ESP-12...

sketch_oct19b.ino

```
}

void setup() {
  Serial.begin(115200);
  pinMode(ledPin, OUTPUT);

  // Status awal LED padam
  digitalWrite(ledPin, LOW);

  connectToWiFi();

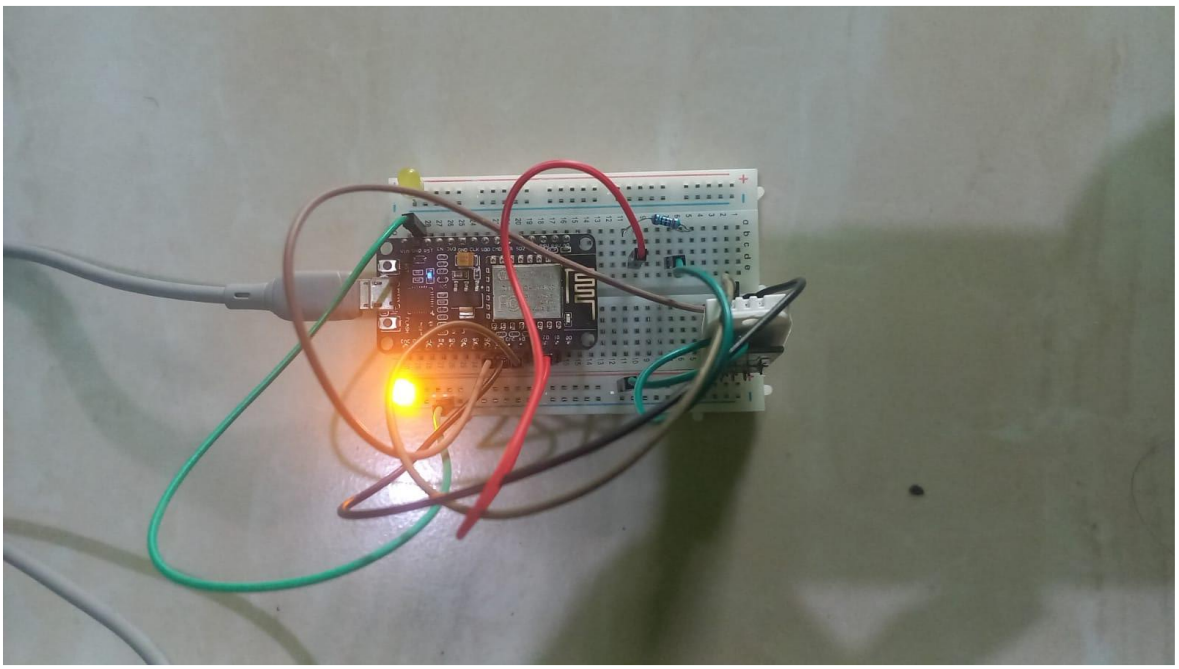
  client.setServer(mqttServer, mqttPort);
  client.setCallback(onMessageReceived); // Daftarkan fungsi callback
}

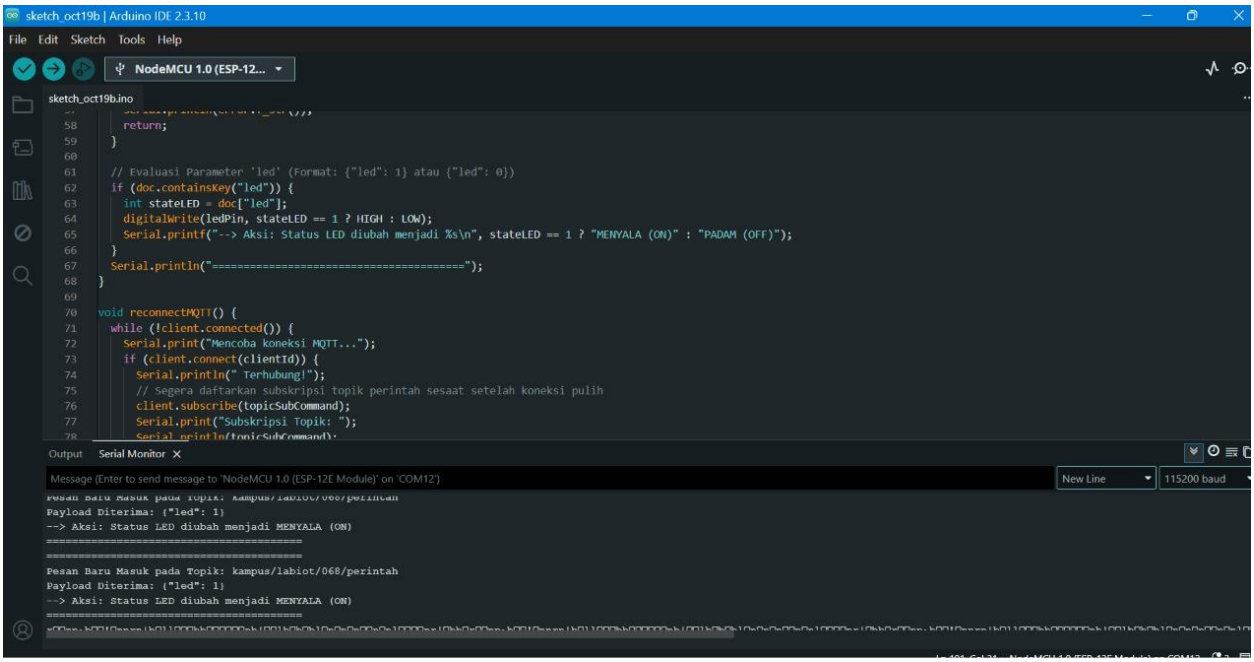
void loop() {
  if (!client.connected()) {
    reconnectMQTT();
  }
  client.loop(); // Wajib untuk terus mendengarkan perintah masuk
}
```

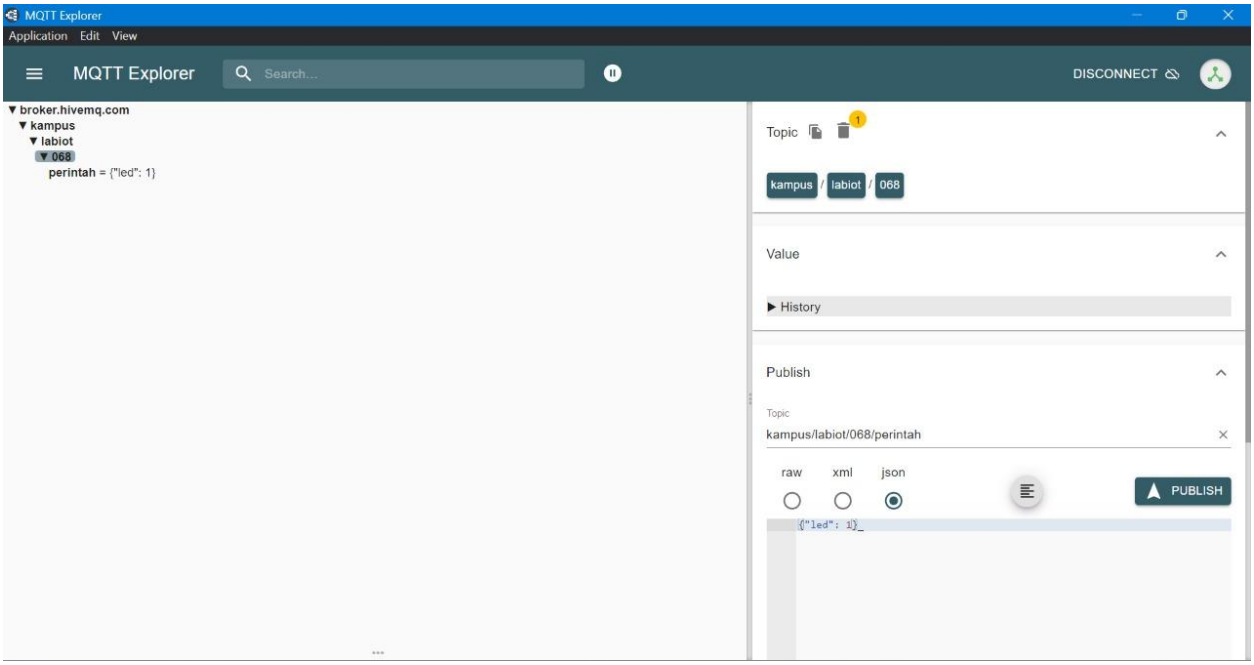
Output Serial Monitor

Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM12')

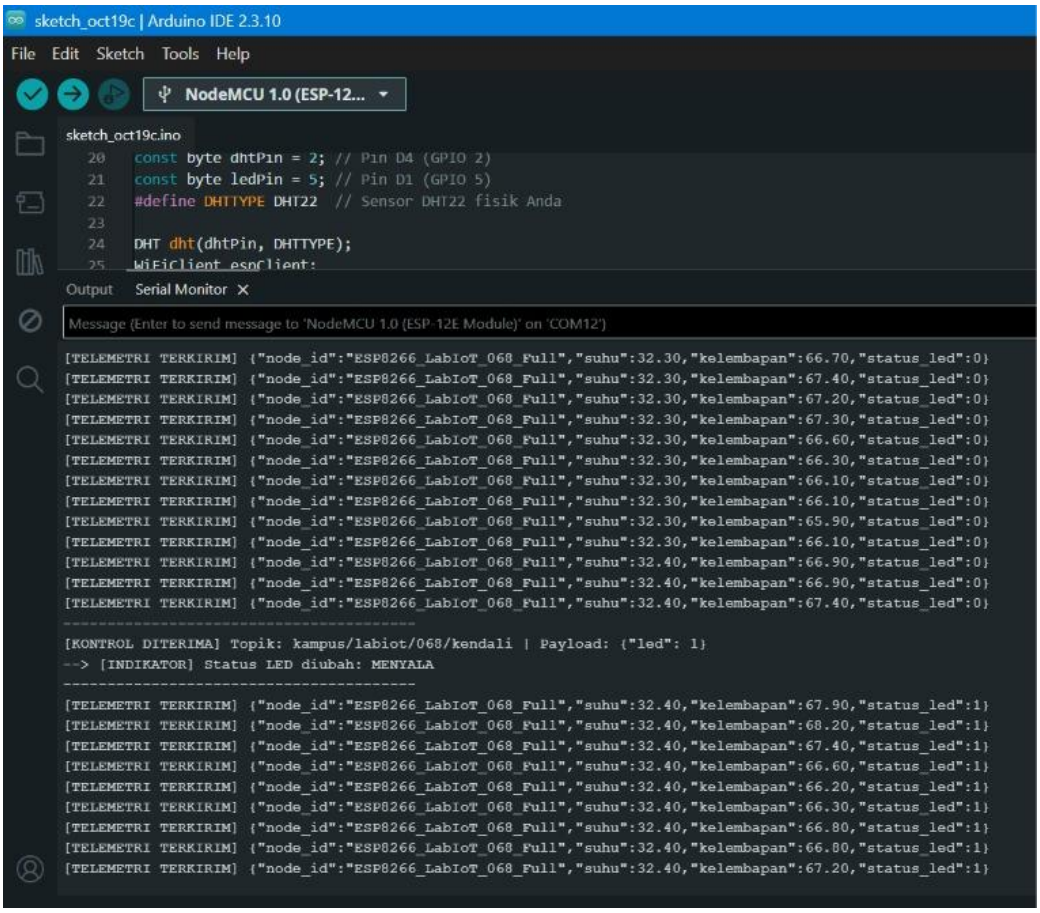
```
...
[OK] Wi-Fi Terkoneksi!
Alamat IP Lokal: 192.168.1.25
Mencoba koneksi MQTT... Terhubung!
Subskripsi Topik: kampus/labiot/068/perintah
```

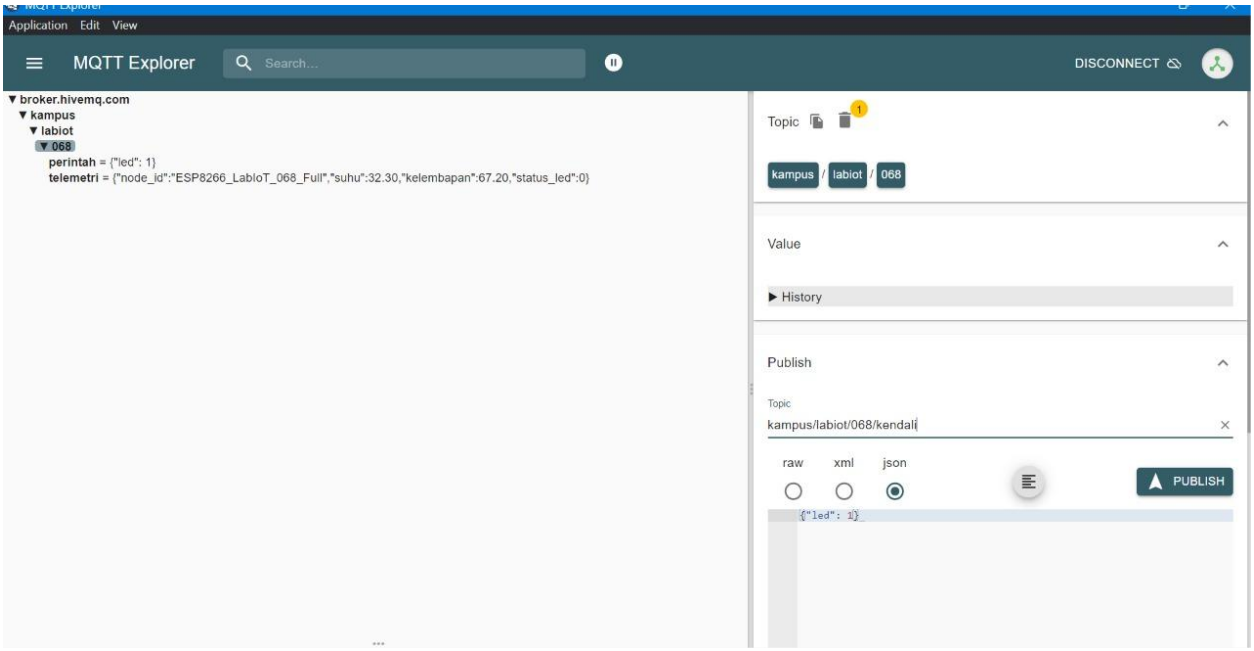


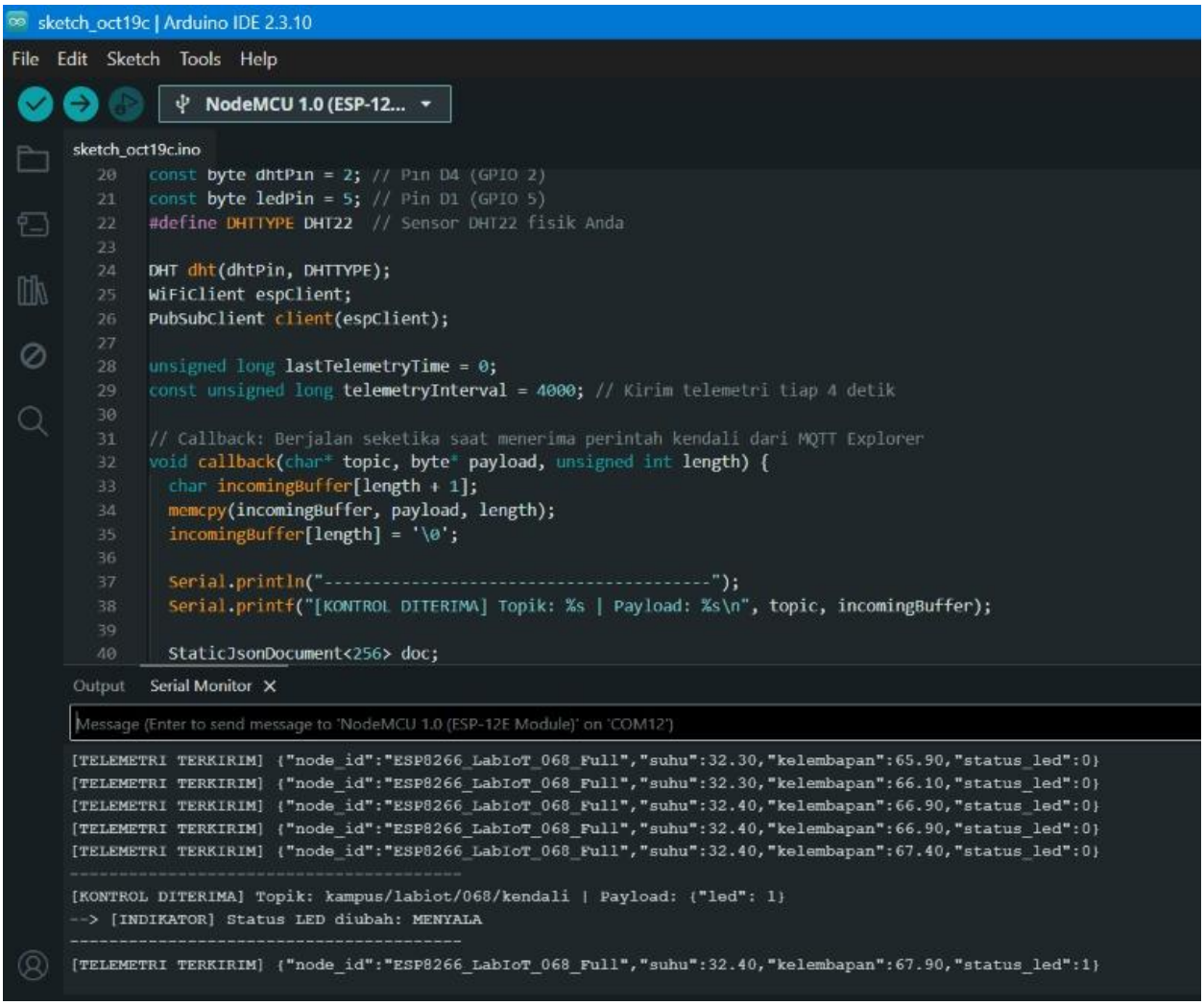

sketch_oct19b | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
sketch_oct19b.ino
return;
}
// Evaluasi Parameter 'led' (Format: {"led": 1} atau {"led": 0})
if (doc.containsKey("led")) {
int stateLED = doc["led"];
digitalWrite(ledPin, stateLED == 1 ? HIGH : LOW);
Serial.printf("--> Aksi: Status LED diubah menjadi %s\n", stateLED == 1 ? "MENYALA (ON)" : "PADAM (OFF)");
}
Serial.println("==");
}
void reconnectMQTT() {
while (!client.connected()) {
Serial.print("Mencoba koneksi MQTT...");
if (client.connect(clientId)) {
Serial.println(" Terhubung!");
// Segera daftarkan subskripsi topik perintah sesaat setelah koneksi pulih
client.subscribe(topicSubCommand);
Serial.print("Subskripsi Topik: ");
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM12')
New Line
115200 baud
Payload Diterima: {"led": 1}
--> Aksi: Status LED diubah menjadi MENYALA (ON)
Pesan Baru Masuk pada Topik: kampus/labiot/068/perintah
Payload Diterima: {"led": 1}
--> Aksi: Status LED diubah menjadi MENYALA (ON)


MQTT Explorer
Application Edit View
MQTT Explorer
Search...
DISCONNECT
broker.hivemq.com
kampus
labiot
068
perintah = {"led": 1}
Topic
kampus / labiot / 068
Value
History
Publish
Topic
kampus/labiot/068/perintah
raw xml json
PUBLISH
{"led": 1}

Prakikum 3


sketch_oct19c | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
sketch_oct19c.ino
const byte dhtPin = 2; // Pin D4 (GPIO 2)
const byte ledPin = 5; // Pin D1 (GPIO 5)
#define DHTTYPE DHT22 // Sensor DHT22 fisik Anda
DHT dht(dhtPin, DHTTYPE);
WiFiClient espClient;
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM12')
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.30,"kelembapan":66.70,"status_led":0}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.30,"kelembapan":67.40,"status_led":0}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.30,"kelembapan":67.20,"status_led":0}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.30,"kelembapan":67.30,"status_led":0}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.30,"kelembapan":66.60,"status_led":0}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.30,"kelembapan":66.30,"status_led":0}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.30,"kelembapan":66.10,"status_led":0}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.30,"kelembapan":66.10,"status_led":0}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.30,"kelembapan":65.90,"status_led":0}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.30,"kelembapan":66.10,"status_led":0}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.40,"kelembapan":66.90,"status_led":0}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.40,"kelembapan":66.90,"status_led":0}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.40,"kelembapan":67.40,"status_led":0}
[KONTROL DITERIMA] Topik: kampus/labiot/068/kendali | Payload: {"led": 1}
--> [INDIKATOR] Status LED diubah: MENYALA
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.40,"kelembapan":67.90,"status_led":1}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.40,"kelembapan":68.20,"status_led":1}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.40,"kelembapan":67.40,"status_led":1}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.40,"kelembapan":66.60,"status_led":1}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.40,"kelembapan":66.20,"status_led":1}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.40,"kelembapan":66.30,"status_led":1}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.40,"kelembapan":66.80,"status_led":1}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.40,"kelembapan":66.80,"status_led":1}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.40,"kelembapan":67.20,"status_led":1}


Application Edit View
MQTT Explorer
Search...
DISCONNECT
broker.hivemq.com
kampus
labiot
068
perintah = {"led": 1}
telemetri = {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.30,"kelembapan":67.20,"status_led":0}
Topic
kampus / labiot / 068
Value
History
Publish
Topic
kampus/labiot/068/kendali
raw xml json
PUBLISH
{"led": 1}


sketch_oct19c | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
sketch_oct19c.ino
const byte dhtPin = 2; // Pin D4 (GPIO 2)
const byte ledPin = 5; // Pin D1 (GPIO 5)
#define DHTTYPE DHT22 // Sensor DHT22 fisik Anda
DHT dht(dhtPin, DHTTYPE);
WiFiClient espClient;
PubSubClient client(espClient);
unsigned long lastTelemetryTime = 0;
const unsigned long telemetryInterval = 4000; // Kirim telemetri tiap 4 detik
// Callback: Berjalan seketika saat menerima perintah kendali dari MQTT Explorer
void callback(char* topic, byte* payload, unsigned int length) {
char incomingBuffer[length + 1];
memcpy(incomingBuffer, payload, length);
incomingBuffer[length] = '\0';
Serial.println("----------");
Serial.printf("[KONTROL DITERIMA] Topik: %s | Payload: %s\n", topic, incomingBuffer);
StaticJsonDocument<256> doc;
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM12')
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.30,"kelembapan":65.90,"status_led":0}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.30,"kelembapan":66.10,"status_led":0}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.40,"kelembapan":66.90,"status_led":0}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.40,"kelembapan":66.90,"status_led":0}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.40,"kelembapan":67.40,"status_led":0}

[KONTROL DITERIMA] Topik: kampus/labiot/068/kendali | Payload: {"led": 1}
--> [INDIKATOR] Status LED diubah: MENYALA

[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_068_Full","suhu":32.40,"kelembapan":67.90,"status_led":1}

Latihan 1 (Analisis Arsitektur & Perbandingan Protokol):
Pada Modul 3, kita menggunakan protokol HTTP untuk membaca dan mengontrol perangkat. Jelaskan secara mendalam mengapa protokol MQTT jauh lebih unggul daripada HTTP dalam hal:

- **Efisiensi Penggunaan Kapasitas Bandwidth pada Paket Data Seluler**
  MQTT jauh lebih hemat penggunaan bandwidth dibandingkan HTTP karena memiliki ukuran *overhead header* yang sangat kecil, yaitu minimal 2 byte per paket, sedangkan HTTP memiliki ukuran header teks berkisar antara 200 hingga 800 byte untuk setiap permintaan. Selain itu, MQTT mempertahankan satu koneksi TCP terbuka secara terus-menerus (*persistent connection*), sehingga transmisi data berkala tidak memerlukan proses jabat tangan (*TCP 3-way handshake*) berulang kali seperti pada protokol HTTP.
- **Beban Komputasi Mikrokontroler (*CPU Overhead*)**
  MQTT berbasis protokol biner yang ringkas dan terstruktur sehingga proses pembacaan (*parsing*) paket data pada level mikrokontroler membutuhkan siklus instruksi CPU dan alokasi memori SRAM yang sangat rendah. Sebaliknya, HTTP berbasis teks panjang yang mengharuskan mikrokontroler mengalokasikan sumber daya komputasi besar untuk membedah string header teks, baris permintaan, dan kode status respons.
- **Kemampuan Pengiriman Data Secara Simultan (*1-to-N Distribution*)**
  MQTT mengadopsi arsitektur *Publish-Subscribe* yang dimediasi oleh perantara cerdas (*MQTT Broker*). Mikrokontroler pengirim hanya perlu memublikasikan paket data satu kali ke satu topik, lalu broker yang akan menggandakan dan menyalurkan data tersebut secara simultan ke banyak penerima (*subscribers*). Pada HTTP, transmisi ke banyak penerima harus dilakukan secara sekuensial satu per satu (*unicast*), yang membebani soket jaringan dan memori mikrokontroler.

Latihan 2 (Analisis Hierarki Topik & Karakter Wildcard):
Sebuah pabrik manufaktur memiliki 3 lini produksi (*line1*, *line2*, *line3*), di mana setiap lini memiliki 4 mesin (*m1*, *m2*, *m3*, *m4*), dan setiap mesin melaporkan metrik suhu, getaran, dan status. Struktur topiknya adalah pabrik/<line>/<mesin>/<sensor>. Tentukan ekspresi topik subskripsi (*subscription topic*) yang tepat menggunakan karakter *wildcard* (+ atau #) untuk:

Struktur topik: pabrik/<line>/<mesin>/<sensor>

1. Topik untuk memantau parameter suhu dari seluruh mesin di semua lini produksi:

pabrik/+/+/suhu

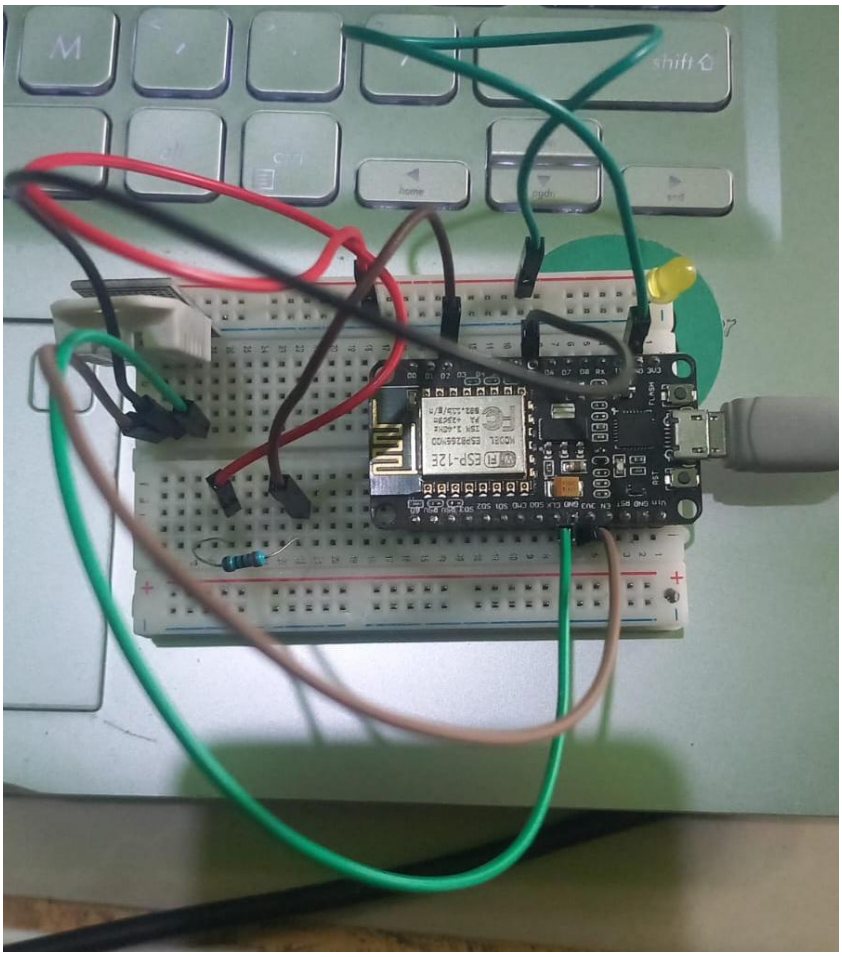
2. Topik untuk memantau seluruh parameter sensor (suhu, getaran, status) khusus untuk mesin m1 di lini line2:

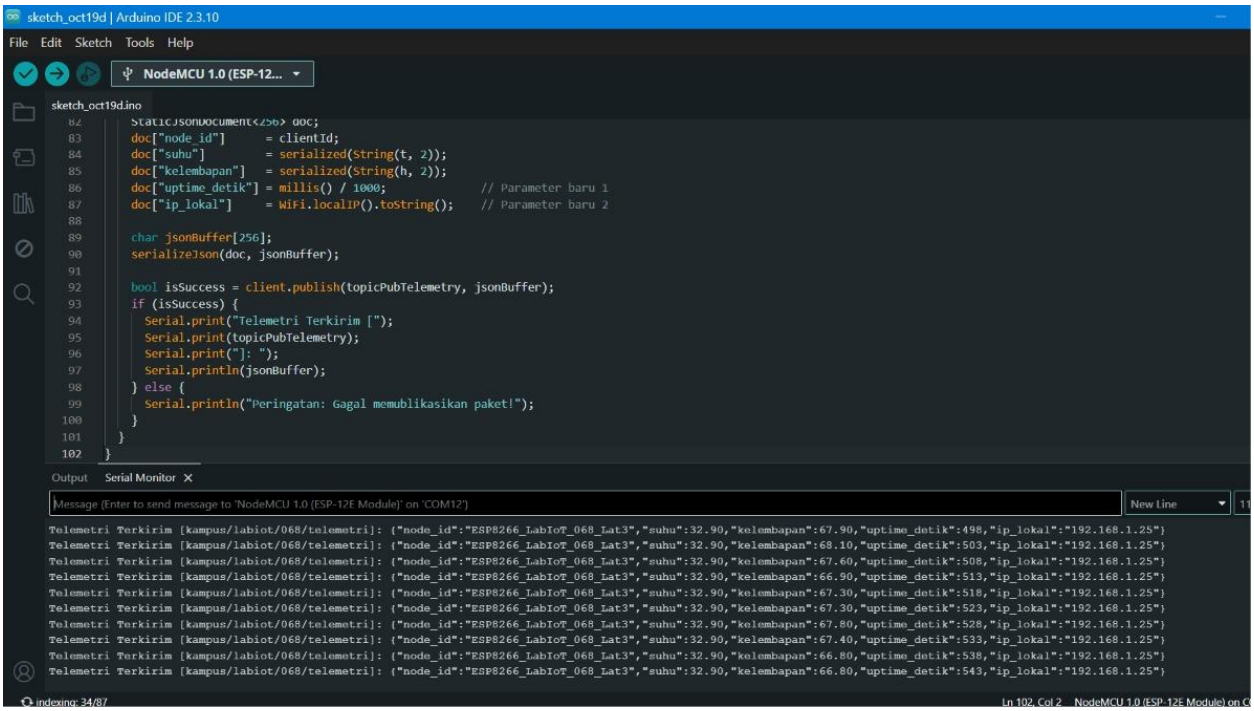
pabrik/line2/m1/#

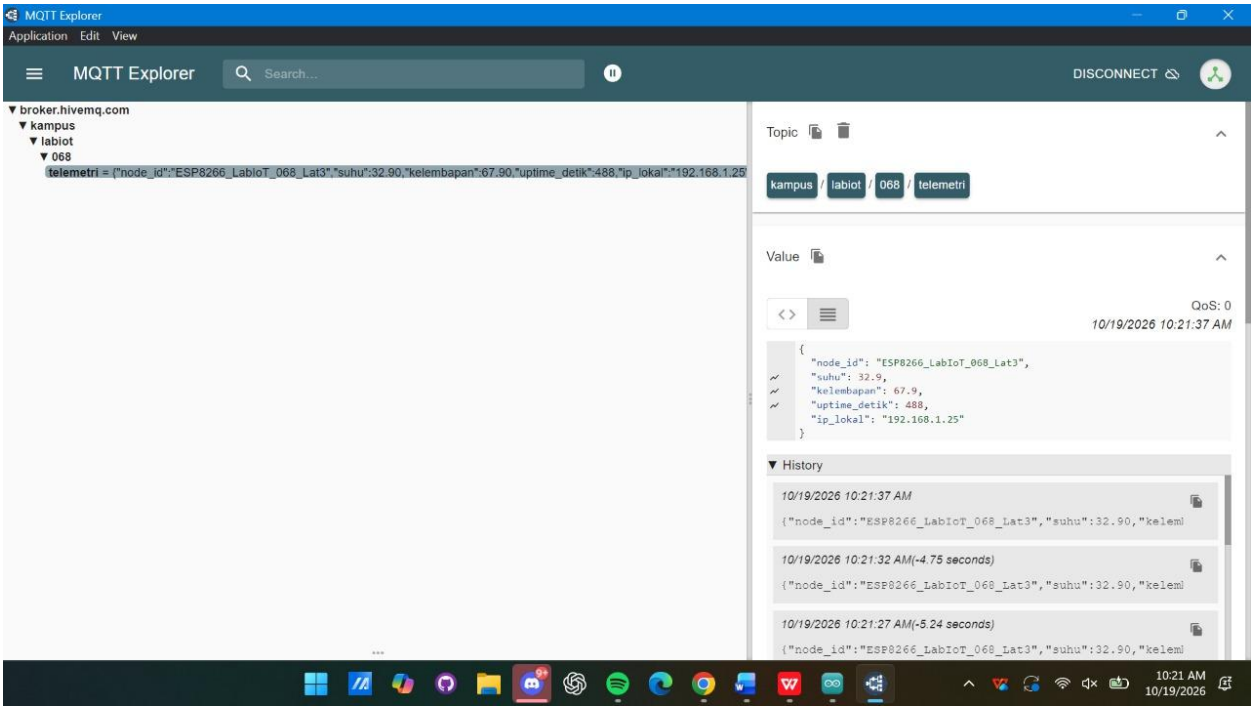
3. Topik untuk memantau seluruh lalu lintas data tanpa terkecuali dari seluruh lini dan seluruh mesin:

pabrik/#

Latihan 3 (Implementasi Serialisasi Data Dinamis):
Modifikasi blok penyusunan JSON pada Praktikum 1 agar paket telemetri menyertakan dua parameter tambahan berikut:



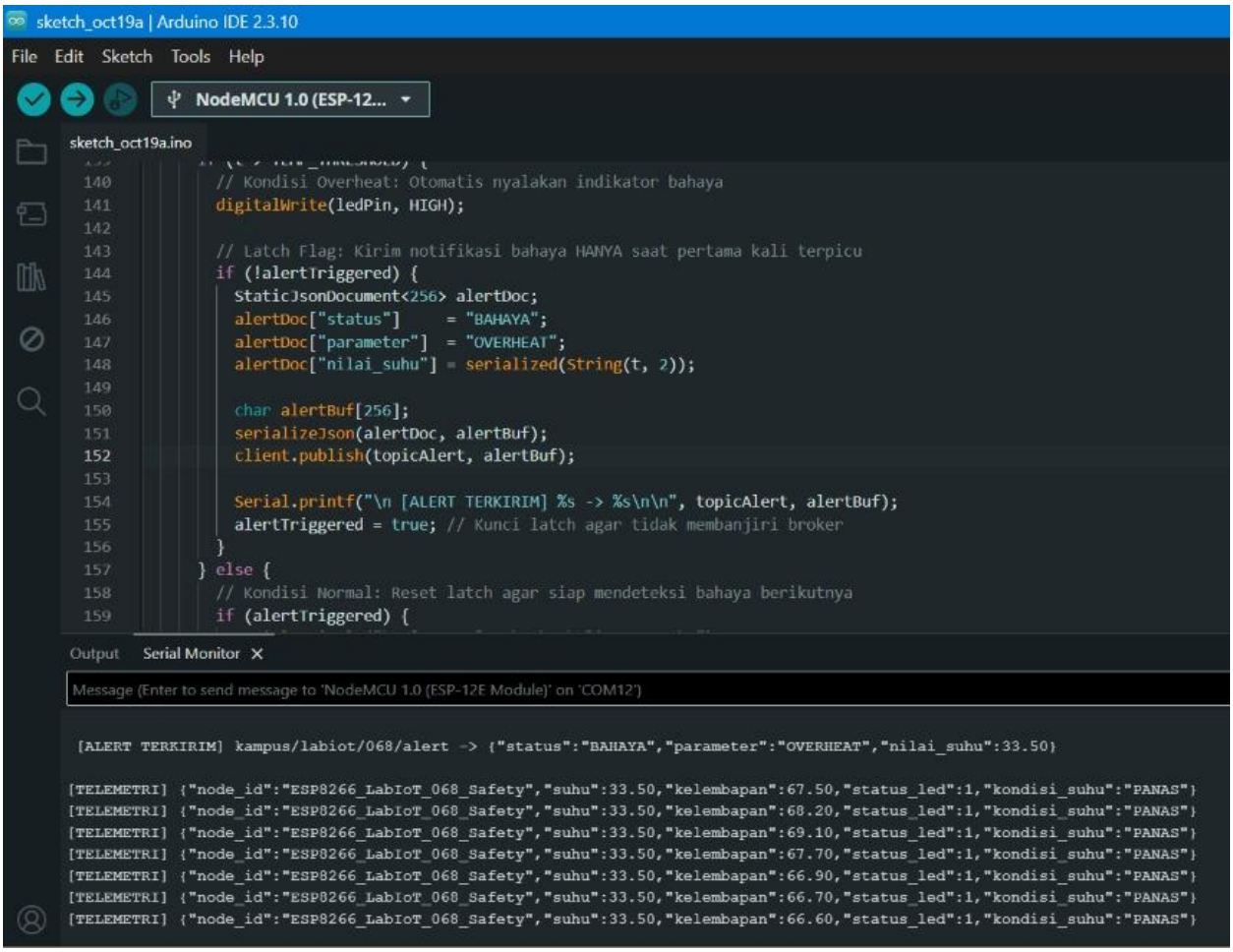

sketch_oct19d | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
sketch_oct19d.ino
StaticJsonDocument<256> doc;
doc["node_id"] = clientId;
doc["suhu"] = serialized(String(t, 2));
doc["kelembapan"] = serialized(String(h, 2));
doc["uptime_detik"] = millis() / 1000; // Parameter baru 1
doc["ip_lokal"] = WiFi.localIP().toString(); // Parameter baru 2
char jsonBuffer[256];
serializeJson(doc, jsonBuffer);
bool isSuccess = client.publish(topicPubTelemetry, jsonBuffer);
if (isSuccess) {
Serial.print("Telemetri Terkirim [");
Serial.print(topicPubTelemetry);
Serial.print("]: ");
Serial.println(jsonBuffer);
} else {
Serial.println("Peringatan: Gagal memublikasikan paket!");
}
}
}
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM12')
New Line
Telemetri Terkirim [kampus/labiot/068/telemetri]: {"node_id":"ESP8266_LabIoT_068_Lat3","suhu":32.90,"kelembapan":67.90,"uptime_detik":498,"ip_lokal":"192.168.1.25"}
Telemetri Terkirim [kampus/labiot/068/telemetri]: {"node_id":"ESP8266_LabIoT_068_Lat3","suhu":32.90,"kelembapan":68.10,"uptime_detik":503,"ip_lokal":"192.168.1.25"}
Telemetri Terkirim [kampus/labiot/068/telemetri]: {"node_id":"ESP8266_LabIoT_068_Lat3","suhu":32.90,"kelembapan":67.60,"uptime_detik":508,"ip_lokal":"192.168.1.25"}
Telemetri Terkirim [kampus/labiot/068/telemetri]: {"node_id":"ESP8266_LabIoT_068_Lat3","suhu":32.90,"kelembapan":66.90,"uptime_detik":513,"ip_lokal":"192.168.1.25"}
Telemetri Terkirim [kampus/labiot/068/telemetri]: {"node_id":"ESP8266_LabIoT_068_Lat3","suhu":32.90,"kelembapan":67.30,"uptime_detik":518,"ip_lokal":"192.168.1.25"}
Telemetri Terkirim [kampus/labiot/068/telemetri]: {"node_id":"ESP8266_LabIoT_068_Lat3","suhu":32.90,"kelembapan":67.30,"uptime_detik":523,"ip_lokal":"192.168.1.25"}
Telemetri Terkirim [kampus/labiot/068/telemetri]: {"node_id":"ESP8266_LabIoT_068_Lat3","suhu":32.90,"kelembapan":67.80,"uptime_detik":528,"ip_lokal":"192.168.1.25"}
Telemetri Terkirim [kampus/labiot/068/telemetri]: {"node_id":"ESP8266_LabIoT_068_Lat3","suhu":32.90,"kelembapan":67.40,"uptime_detik":533,"ip_lokal":"192.168.1.25"}
Telemetri Terkirim [kampus/labiot/068/telemetri]: {"node_id":"ESP8266_LabIoT_068_Lat3","suhu":32.90,"kelembapan":66.80,"uptime_detik":538,"ip_lokal":"192.168.1.25"}
Telemetri Terkirim [kampus/labiot/068/telemetri]: {"node_id":"ESP8266_LabIoT_068_Lat3","suhu":32.90,"kelembapan":66.80,"uptime_detik":543,"ip_lokal":"192.168.1.25"}
indexing: 34/87
Ln 102, Col 2 NodeMCU 1.0 (ESP-12E Module) on C


MQTT Explorer
Application Edit View
MQTT Explorer
Search...
DISCONNECT
broker.hivemq.com
kampus
labiot
068
telemetri = {"node_id":"ESP8266_LabIoT_068_Lat3","suhu":32.90,"kelembapan":67.90,"uptime_detik":488,"ip_lokal":"192.168.1.25
Topic
kampus / labiot / 068 / telemetri
Value
QoS: 0
10/19/2026 10:21:37 AM
{
"node_id": "ESP8266_LabIoT_068_Lat3",
"suhu": 32.9,
"kelembapan": 67.9,
"uptime_detik": 488,
"ip_lokal": "192.168.1.25"
}
History
10/19/2026 10:21:37 AM
{"node_id":"ESP8266_LabIoT_068_Lat3","suhu":32.90,"kelem
10/19/2026 10:21:32 AM(-4.75 seconds)
{"node_id":"ESP8266_LabIoT_068_Lat3","suhu":32.90,"kelem
10/19/2026 10:21:27 AM(-5.24 seconds)
{"node_id":"ESP8266_LabIoT_068_Lat3","suhu":32.90,"kelem
10:21 AM
10/19/2026

Latihan 4 (Analisis Mekanisme Reconnect & Loop Blocking):
Pada Praktikum 3, di dalam fungsi loop() terdapat pemanggilan rutin client.loop().

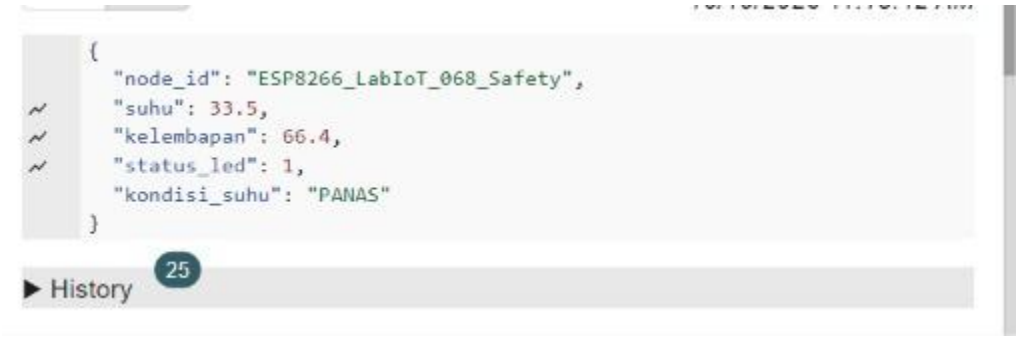
- Konsekuensi Teknis Penyisipan Instruksi delay(6000) di Dalam loop()
  Instruksi delay(6000) bersifat *blocking* dan menghentikan seluruh siklus instruksi mikrokontroler selama 6 detik. Hal ini menyebabkan fungsi pemeliharaan jaringan client.loop() terlambat dieksekusi, sehingga fungsi *callback* penerimaan perintah kontrol LED mengalami latensi tinggi (tidak responsif). Selain itu, jika durasi delay melewati ambang batas waktu *keep-alive ping*, broker MQTT akan menganggap klien tidak aktif dan memutuskan sambungan TCP (*timeout*).
- Cara Kerja Fungsi reconnect() dalam Memulihkan Koneksi
  Fungsi reconnect() bekerja dengan memeriksa status koneksi menggunakan perulangan while (!client.connected()). Selama mikrokontroler terputus dari broker, sistem akan mencoba menyambung kembali dengan memanggil instruksi client.connect(). Sesaat setelah sambungan berhasil terhubung kembali, fungsi ini segera memanggil client.subscribe() untuk mendaftarkan ulang topik perintah kontrol, karena data langganan topik pada broker terhapus secara otomatis saat koneksi TCP terputus.

Tugas: Sistem Telemetri & Kendali Closed-Loop Industri





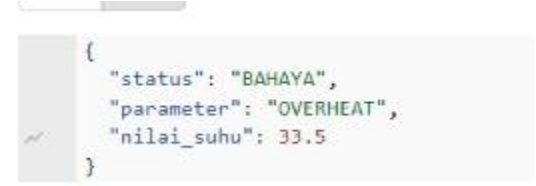
Percobaan 1



Percobaan 2



Percobaan 3



Percobaan 4



